

Math 104 Homework 6

Due Date: Monday, August 10 at 11:59pm

Instructions: Submit pdf solutions in LaTeX (preferred), or *neatly* handwritten. Unless otherwise stated, you may cite any result from class or from Rudin's book, but all other assertions you make must be proved rigorously. At the top of your submission, please write down the names of the other students you collaborated with and any outside sources you consulted.

1. Let $a \in \mathbb{R}$, and suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is Riemann integrable on $[a, b]$ for all $b > a$. We define

$$\int_a^\infty f(x) dx = \lim_{b \rightarrow \infty} \int_a^b f(x) dx.$$

if the limit exists. We call this an *improper integral* of f . We say the improper integral *converges* if this limit exists and *diverges* otherwise.

- (a) Suppose $\int_a^\infty |f(x)| dx$ converges. Prove that $\int_a^\infty f(x) dx$ also converges (this is called *absolute convergence*).
- (b) Suppose further that $a = 1$ and $f(x)$ is nonnegative and monotonically decreasing on $[1, \infty)$. Prove that

$$\int_1^\infty f(x) dx \quad \text{converges if and only if} \quad \sum_{n=1}^\infty f(n) \quad \text{converges}.$$

This is the *integral test*. *Hint:* draw a picture.

- (c) Find an example of a function $g : [1, \infty) \rightarrow \mathbb{R}$, integrable on $[1, b]$ for all $b > 1$, so that

$$\int_1^\infty g(x) dx$$

is convergent but not absolutely convergent. Prove your assertions.

2. Let (X, d_X) be a metric space. Suppose $\{f_n\}$ and $\{g_n\}$ are sequences of functions $X \rightarrow \mathbb{R}$ that converge uniformly to f and g respectively.

- (a) Show that the sequence $\{f_n + g_n\}$ converges uniformly to $f + g$.

- (b) Suppose further that, for every $n \in \mathbb{N}$, the functions f_n and g_n are bounded.
- Show that f and g (and hence fg) are bounded.
 - Show that the sequences $\{f_n\}$ and $\{g_n\}$ are *uniformly bounded*, in that there is $M > 0$ so that $|f_n(x)| \leq M$ and $|g_n(x)| \leq M$ for all $n \in \mathbb{N}$ and for all $x \in X$.
 - Show that $f_n g_n \rightarrow fg$ uniformly.
- (c) If we drop the assumption that every f_n is bounded (but keep the assumptions that every g_n is bounded and $f_n \rightarrow f$ and $g_n \rightarrow g$ uniformly), is it still true that $f_n g_n \rightarrow fg$ uniformly? Prove or give a counterexample.
3. The purpose of this problem is to define the natural logarithm. You are not allowed to use any properties of this function you might know already (i.e. do not take anything for granted). However, you are allowed to use all of the facts we proved about the exponential function in class.

Define a function $f : (0, \infty) \rightarrow \mathbb{R}$ via

$$f(x) = \int_1^x \frac{1}{t} dt.$$

- (a) Using the change of variables theorem, prove that

$$f(ab) = f(a) + f(b)$$

for all $a, b > 0$.

- (b) Prove that

$$\lim_{x \rightarrow \infty} f(x) = \infty.$$

- (c) Prove that f is differentiable with

$$f'(x) = \frac{1}{x}$$

for all $x \in (0, \infty)$.

- (d) Prove that $f(e^x) = x$ for all $x \in \mathbb{R}$. *Hint:* differentiate both sides.

- (e) Prove that $e^{f(x)} = x$ for all $x \in (0, \infty)$ *Hint:* differentiate $\frac{e^{f(x)}}{x}$.

(f) Prove that

$$f(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} (x-1)^n$$

for all $x \in (0, 2)$. *Hint:* write $\frac{1}{x}$ as a geometric series.

(g) Prove that

$$f(2) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}.$$

Hint: you cannot just plug 2 into part (e) as we do not yet know whether the formula from (e) is valid for $x = 2$ (or whether this series even converges). There is a theorem that will help you here.

4. For $\alpha \in \mathbb{R}$ and $k \in \{0, 1, 2, \dots\}$, define the *generalized binomial coefficient*

$$\binom{\alpha}{k} = \frac{\alpha(\alpha-1)(\alpha-2)\cdots(\alpha-k+1)}{k!}.$$

By convention, we always have $\binom{\alpha}{0} = 1$. The *binomial series* is the power series

$$\sum_{k=0}^{\infty} \binom{\alpha}{k} x^k.$$

(a) If $\alpha \in \{0, 1, 2, \dots\}$, show that the binomial series converges to $(1+x)^\alpha$ for all $x \in \mathbb{R}$.

(b) If $\alpha \notin \{0, 1, 2, \dots\}$ (which we will assume for the rest of this problem), show that the binomial series has radius of convergence 1. *Hint:* Ratio test.

(c) Let $f : (-1, 1) \rightarrow \mathbb{R}$ be defined by $f(x) = \sum_{k=0}^{\infty} \binom{\alpha}{k} x^k$. Show that f is differentiable and satisfies

$$(1+x)f'(x) = \alpha f(x)$$

for all $x \in (-1, 1)$.

(d) Show that $f(x) = (1+x)^\alpha$ for all $x \in (-1, 1)$ *Hint:* differentiate the function $g(x) = (1+x)^{-\alpha} f(x)$.

(e) Show that

$$\sqrt{2} = \sum_{k=0}^{\infty} \binom{1/2}{k}.$$

Hint: you cannot just plug in 1 into the binomial series; the same comments from problem 3 part (f) apply.